WATER PURIFICATION

R26-IT-005



SLIIT
FACULTY OF COMPUTING

2026/03/16

PROBLEM



Despite technological advances, safe drinking water is a global challenge due to urbanization, industrial discharge, and climate change.

In Sri Lanka, the situation is critical: contaminated groundwater with fluoride and heavy metals is linked to Chronic Kidney Disease of unknown origin.

Current methods are inefficient, waste water/energy, and lack intelligent monitoring.



USERS & STAKEHOLDERS

USERS	STAKEHOLDERS
Urban and rural households	Water supply boards Public health authorities NGOs Environmental agencies



IMPACT OF THE PROBLEM TO USER &SOCIETY



Health - Waterborne diseases and long-term health complications (CKDu).

Economic - Increased healthcare costs and lost productivity.

Environmental - Inefficient water usage and energy waste from existing purifiers.

EXISTING SOLUTIONS & THEIR LIMITATIONS



Solution - Current purification methods like RO systems waste water and strip essential minerals, distillation is energy-intensive and slow, and chlorination risks harmful by-products.

Limitation - They operate in isolation with no real-time monitoring or adaptive control.



PROPOSED SOLUTIONS & LIMITATIONS

Solution - An integrated system combining IoT, AI/ML, and autonomous systems will identify water types, optimize purification, maintain safe mineral balance, and monitor environmental sources in real-time.

Limitation - This intelligent system uses AI to select the optimal purification method based on real-time data, operates efficiently through low-power distillation and precise mineral dosing, and proactively monitors environmental sources to predict contamination.



IT22295460 | W.A.S.P.D. WIJESINGHA

HOUSEHOLD DISTILLED WATER MAKER WITH LEAST POWER – IOT + MOBILE APP

Novelty - The first integration of AI-assisted energy optimization with a household distiller. Unlike passive systems, it learns usage patterns and predicts scaling/faults, transforming a dumb appliance into a smart, sustainable device.



PROBLEM

Conventional home distillers consume massive amounts of electricity and lack user control, making them expensive and unsustainable to run.

SOLUTION

An intelligent, IoT-integrated distillation system that optimizes energy consumption through AI-assisted heating control. It features real-time monitoring of temperature and energy usage via ESP32 sensors, adaptive heating cycles based on usage patterns, and a mobile app for remote control and fault prediction dramatically reducing energy costs while transforming traditional distillers into smart, sustainable appliances.

OBJECTIVES



To design, develop, and validate a smart, energy-optimized household distilled water generation system that integrates IoT-based monitoring and mobile application control to enhance user convenience and operational efficiency.

SUB OBJECTIVES

- Energy-efficient design: Insulated boiling chamber + counter-flow heat exchanger to preheat feed water.
- IoT control: ESP32 monitors temp, water levels; PID/PWM optimizes heating cycles.
- Mobile app: Flutter-based remote control, real-time data, history logs, and alerts.
- Validation: Energy consumption (kWh/L) compared vs. conventional distillers; sensor accuracy and app reliability tested.

METHODOLOGY



Application of Domain Knowledge

- Thermodynamics: Heat recovery via heat exchanger; insulated chambers to reduce loss.
- Embedded Systems: ESP32 with sensors, PID/PWM for precise heating control.
- IoT & Cloud: MQTT/Firebase for real-time data sync and remote access.
- Mobile Dev: Flutter app for monitoring, control, logs, and alerts.
- Data Analytics: Usage analysis for predictive maintenance and energy optimization

Latest Technologies

- Hardware: Peltier elements or low-wattage heating elements, ESP32/Arduino IoT modules.
- Software: MQTT protocol for IoT communication, Flutter/Kotlin for mobile app development, ML for predictive maintenance (anomaly detection).

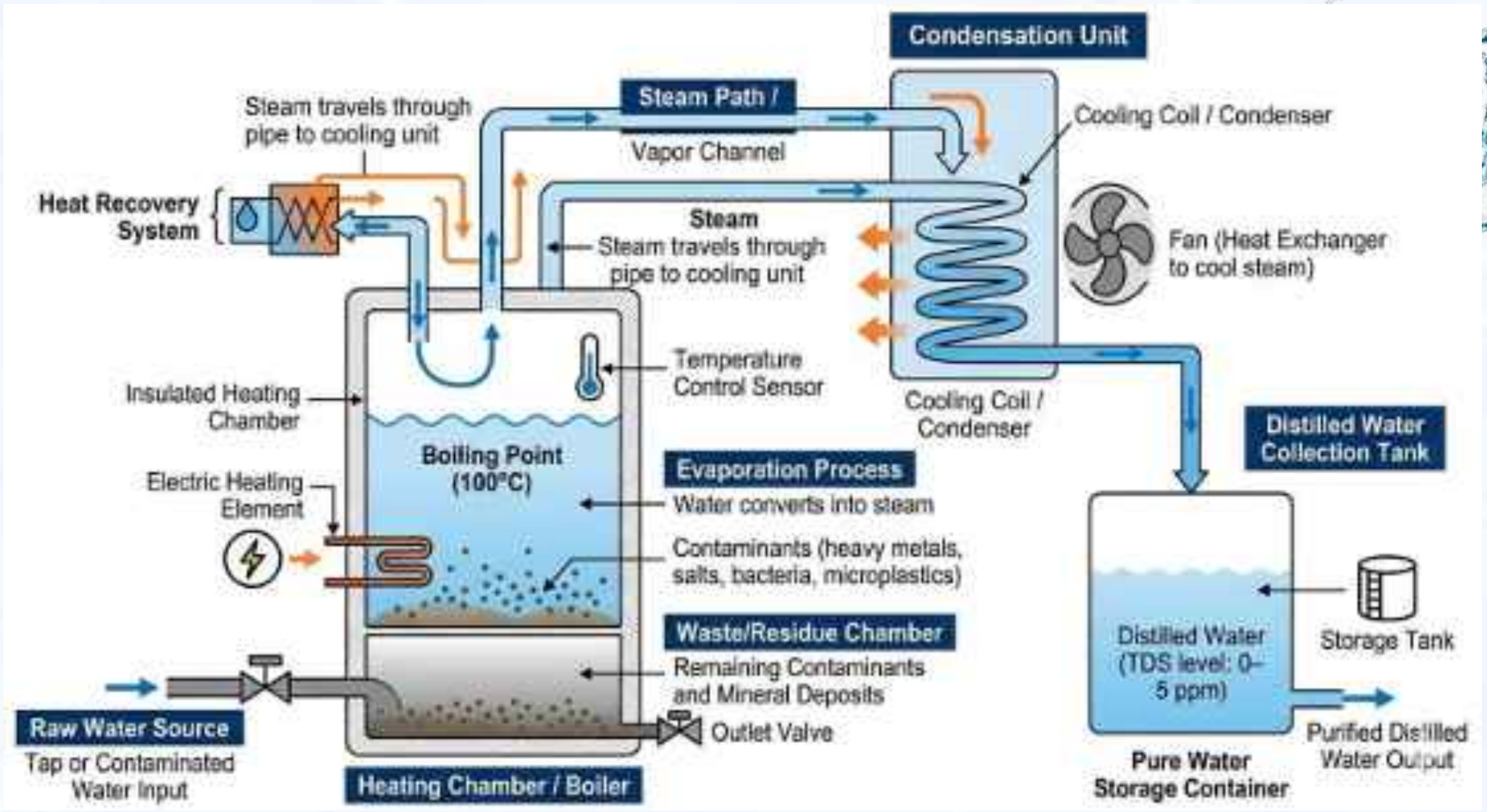
Validation & Evaluation

- Measure energy consumption (kWh per liter) vs. traditional distillers.
- User satisfaction survey for the mobile app interface.
- Reliability testing for fault prediction algorithms.

Data & Ethics

- Data: Energy usage data generated during prototype testing.
- Ethics: N/A for hardware performance, but user data from the app will be anonymized.

SYSTEM DIAGRAM





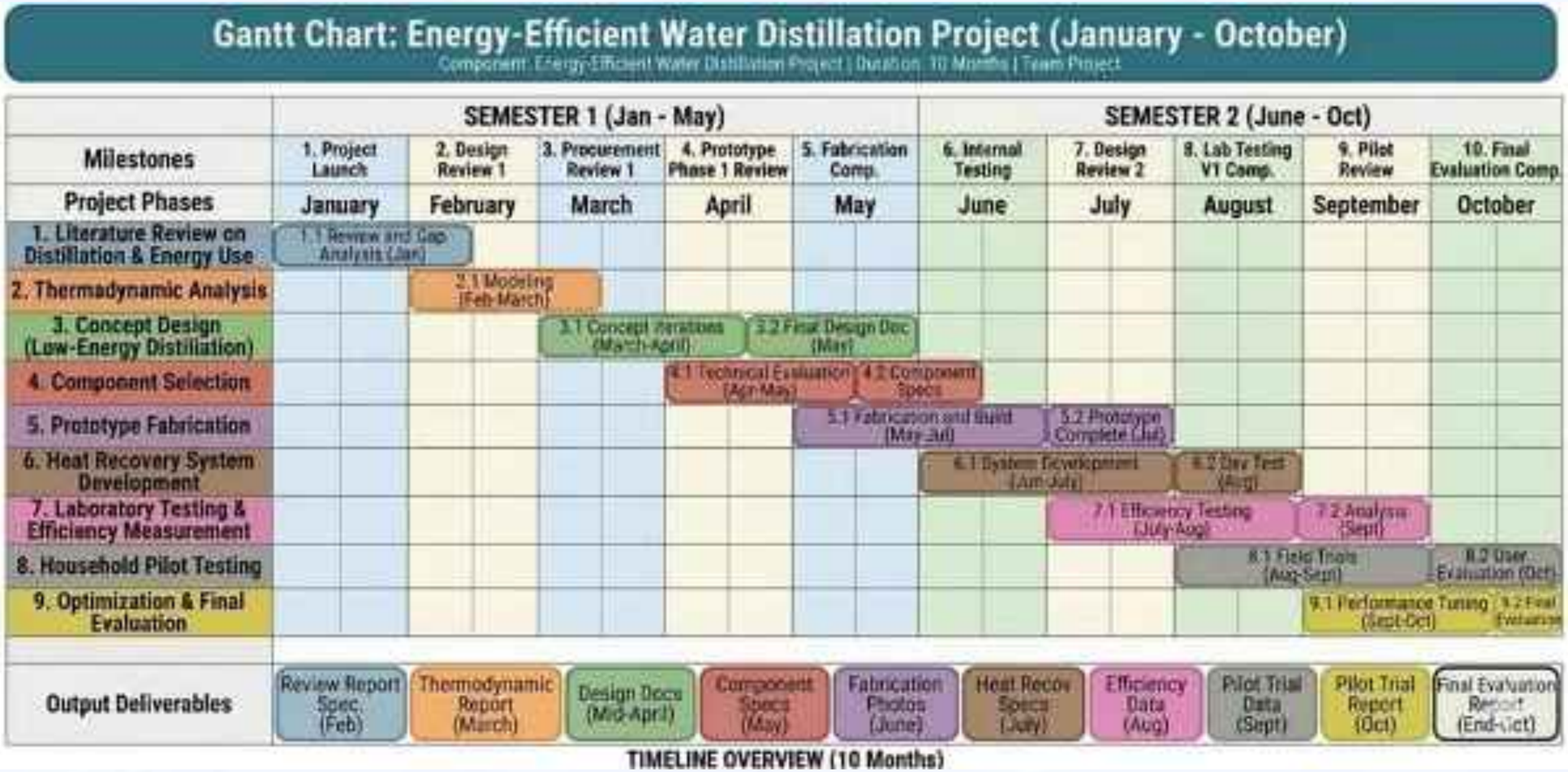
DEPENDENCIES

- Requires stable Wi-Fi for remote monitoring and control.
- Depends on De Silva's water classification for source water quality input.

COMMON COMPONENTS

- Shares cloud backend (Firebase) with other subsystems.
- Common database for energy logs and user data.
- Integrated with overall system dashboard for unified monitoring.

GANTT CHART


Gantt Chart: Energy-Efficient Water Distillation Project (January - October)
Component: Energy-Efficient Water Distillation Project | Duration: 10 Months | Team Project
SEMESTER 1 (Jan - May)
SEMESTER 2 (June - Oct)
Milestones
1. Project Launch
2. Design Review 1
3. Procurement Review 1
4. Prototype Phase 1 Review
5. Fabrication Comp.
6. Internal Testing
7. Design Review 2
8. Lab Testing V1 Comp.
9. Pilot Review
10. Final Evaluation Comp.
Project Phases
January
February
March
April
May
June
July
August
September
October
1. Literature Review on Distillation & Energy Use
1.1 Review and Gap Analysis (Jan)
2. Thermodynamic Analysis
2.1 Modeling (Feb-March)
3. Concept Design (Low-Energy Distillation)
3.1 Concept Iterations (March-April)
3.2 Final Design Doc (May)
4. Component Selection
4.1 Technical Evaluation (Apr-May)
4.2 Component Specs
5. Prototype Fabrication
5.1 Fabrication and Build (May-Jul)
5.2 Prototype Complete (Jul)
6. Heat Recovery System Development
6.1 System Development (Jun-July)
6.2 Dev Test (Aug)
7. Laboratory Testing & Efficiency Measurement
7.1 Efficiency Testing (July-Aug)
7.2 Analysis (Sept)
8. Household Pilot Testing
8.1 Field Trials (Aug-Sept)
8.2 User Evaluation (Oct)
9. Optimization & Final Evaluation
9.1 Performance Tuning (Sept-Oct)
9.2 Final Evaluation
Output Deliverables
Review Report Spec. (Feb)
Thermodynamic Report (March)
Design Docs (Mid-April)
Component Specs (May)
Fabrication Photos (June)
Heat Recov Specs (July)
Efficiency Data (Aug)
Pilot Trial Data (Sept)
Pilot Trial Report (Oct)
Final Evaluation Report (End-Oct)
TIMELINE OVERVIEW (10 Months)



FUNCTIONAL REQUIREMENTS

- THE SYSTEM SHALL PRODUCE DISTILLED WATER WITH A PURITY LEVEL OF LESS THAN 10 PPM TOTAL DISSOLVED SOLIDS (TDS).
- THE SYSTEM SHALL ACHIEVE A MINIMUM OF 30% ENERGY SAVINGS PER LITER OF DISTILLED WATER PRODUCED COMPARED TO A CONVENTIONAL, NON-OPTIMIZED COUNTERTOP DISTILLER OF SIMILAR CAPACITY. THIS WILL BE ACHIEVED THROUGH HEAT RECOVERY AND PID-BASED HEATING CONTROL.

NON-FUNCTIONAL REQUIREMENTS

- THE MOBILE APPLICATION MUST HAVE AN INTUITIVE INTERFACE WITH CLEAR DATA VISUALIZATION (E.G., GAUGES, SIMPLE ICONS) THAT REQUIRES MINIMAL TRAINING FOR A USER TO OPERATE EFFECTIVELY.
- THE SYSTEM SHALL BE CAPABLE OF OPERATING CONTINUOUSLY FOR AT LEAST ONE FULL DISTILLATION CYCLE (3-4 HOURS) WITHOUT ANY WI-FI COMMUNICATION DROPOUTS, SYSTEM FREEZES, OR UNEXPECTED FAILURES.

BUDGET & JUSTIFICATION

Item	Component/Description	Estimated Cost (LKR)	Justification
1. Distillation Core	Stainless Steel Boiling Chamber, 750W Heating Element, High-Temp Insulation	3,500	The heart of the system; material choice and Insulation are key for efficiency and safety.
3. Condenser Unit	Copper or Stainless-Steel Condenser Coil, Small Fan (optional)	8,000	For efficiently cooling and condensing the steam.
4. Enclosure & Tanks	Outer Housing (Plastic or Metal), BPA-free Feed and Collection Tanks (2x)	4,000	For structural integrity, aesthetics, and safe water storage.
5. Core Electronics	ESP32 Development Board, Power Supply (12V/5V), Solid State Relay (SSR)	5,500	The "brain" of the system is for control, connectivity, and safe switching of high power.
7. Misc. Hardware	Tubing, Fittings, Waterproof Connectors, Wires, Screws, etc.	4,500	Essential for plumbing and assembling all components.
TOTAL ESTIMATED BUDGET		**25,500**	

REFERENCES



[1] Kim, J., & Lee, H. (2018). Energy-Efficient Distillation Techniques for Household Applications. Energy & Buildings, 160, 122-130. (Key paper on the core topic of efficiency).

[2] Patel, A., & Mehta, S. (2020). IoT-Enabled Smart Water Purification Systems: A Review.Journal of Smart Technologies, 8(3), 120-132. (Provides context on integrating IoT into water purification).

[3] Sharma, R., et al. (2019). Comparative Study of Water Purification Technologies for Domestic Applications. International Journal of Environmental Science and Technology, 14(2), 75-84. (Benchmarks distillation against other methods).

[4] Singh, P., et al. (2021). IoT-Based Monitoring and Control for Smart Home Appliances: A Comprehensive Review. IEEE Access, 9, 89012-89025. (Broad overview of IoT in home appliances, including design patterns).

[5] World Health Organization (WHO). (2017).Guidelines for Drinking-water Quality (4th ed.). WHO. (For context on water purity standards).



IT22212122 | D. DODANWELA

INTELLIGENT ALKALINE WATER GENERATION MODULE – IOT & AI/ML

Novelty - Replaces "dumb" cartridges with a closed-loop, AI-driven dosing system. It's the first system to dynamically balance minerals based on real-time feedback, ensuring consistent, safe, and personalized water quality.



PROBLEM

Existing alkaline water filters use static mineral cartridges that deplete over time and cannot adjust to the incoming water chemistry, risking over-mineralization or ineffective pH change.

SOLUTION

An intelligent, closed-loop dosing system that dynamically adds essential minerals (Calcium, Magnesium, Potassium) to purified water based on real-time sensor feedback. Unlike static cartridges, it uses an AI model to calculate precise mineral dosages, ensuring consistent, safe alkalinity that meets WHO standards while adapting to varying incoming water chemistry.

OBJECTIVES



To design, develop, and validate an intelligent, closed-loop alkaline water generation module that integrates IoT-based real-time sensing with a machine learning-driven control algorithm to automatically and precisely maintain a user-defined, safe, and mineral-balanced alkaline water profile.

SUB OBJECTIVES

- **Senses pre/post-mineralization water parameters (pH, TDS, minerals) via ESP32 and sensors.**
- **Calculates optimal mineral dosages using a trained ML regression model (ANN/SVR).**
- **Doses precisely via peristaltic pumps injecting calcium, magnesium, and potassium.**
- **Controls continuously through ESP32-integrated feedback loop.**
- **Interfaces via mobile/web dashboard for user control and validation of pH accuracy and stability.**

METHODOLOGY



Application of Domain Knowledge

- Biochemistry: Mineral impact on health and safe alkalinity.
- Control Systems: Automated closed-loop dosing.
- Machine Learning: Regression models (TensorFlow) for dosage prediction.
- Sensor Technology: pH, ORP, ion-specific sensors for real-time feedback.
- Actuators: Peristaltic pumps for micro-precision mineral injection.

Latest Technologies

- Sensors: pH, ORP, Mineral-specific ion sensors.
- Actuators: Peristaltic pumps for precise mineral dosing.
- Algorithms: Python/TensorFlow for regression models to predict final pH based on dosage.

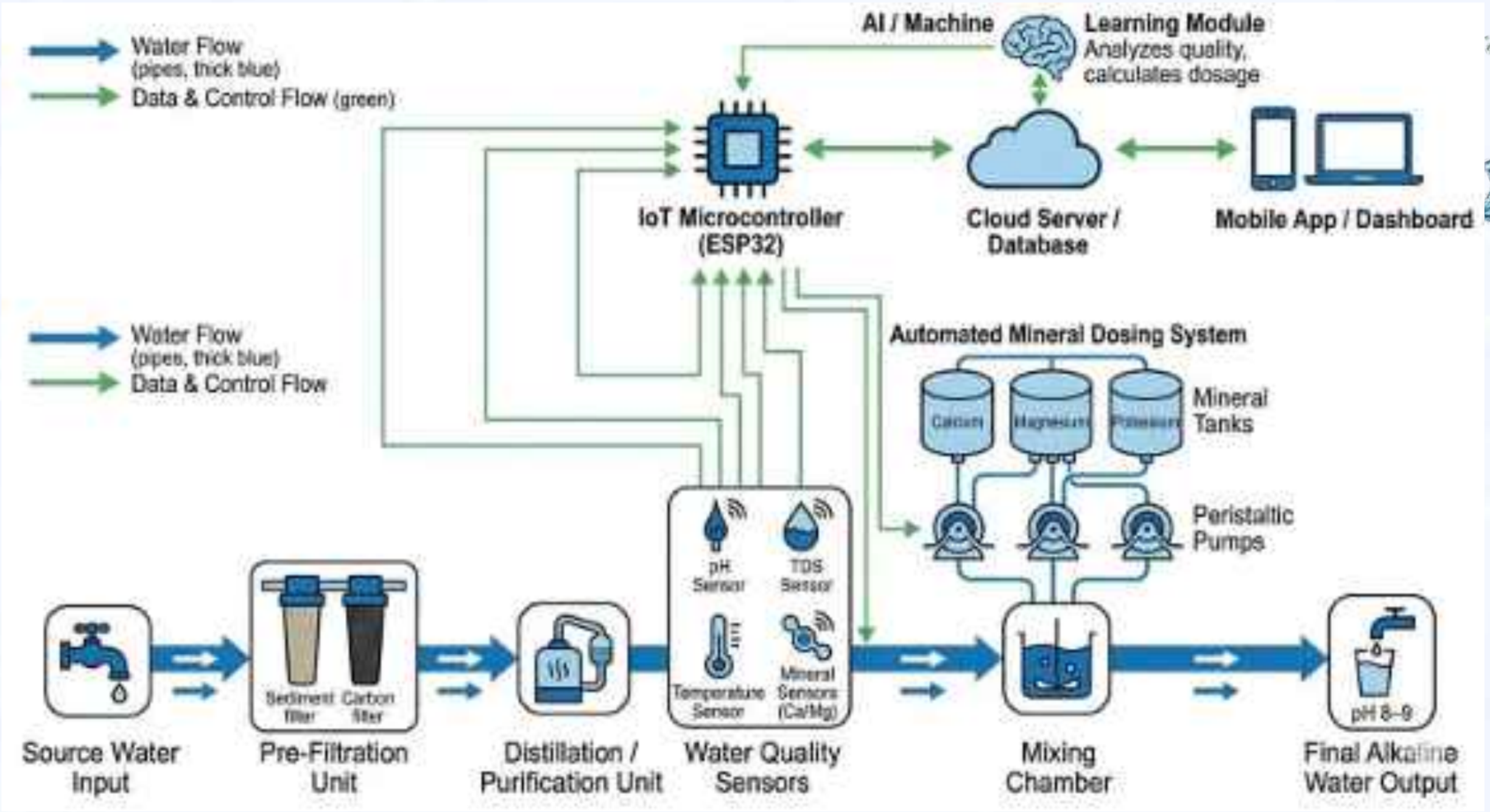
Validation & Evaluation

- Measure pH stability and mineral concentration consistency over time.
- Compare mineral retention vs. standard RO/Alkaline filters.

Data & Ethics

- Data: Feedback loop data from the system's own sensors.
- Ethics: Clearance required if conducting taste tests with human participants, ensuring mineral levels are safe.

SYSTEM DIAGRAM





FUNCTIONAL REQUIREMENTS

- Monitor incoming water pH, TDS, and temperature in real time.
- Monitor output alkaline water quality (pH and TDS).
- Collect sensor data using an ESP32 microcontroller.
- Process water quality data using an AI/ML model.
- Automatically inject Calcium, Magnesium, and Potassium minerals into the water.
- Control mineral dosing using peristaltic pumps.

NON-FUNCTIONAL REQUIREMENTS

- The system must maintain high accuracy in controlling pH levels (±0.15).
- The system should respond to changes in water quality within 2–3 minutes.
- The system must be reliable for continuous operation.
- The system must include safety mechanisms to stop dosing if unsafe conditions occur.
- The system should be easy to maintain, allowing simple sensor calibration and mineral replacement.
- The system should provide an intuitive and user-friendly interface.

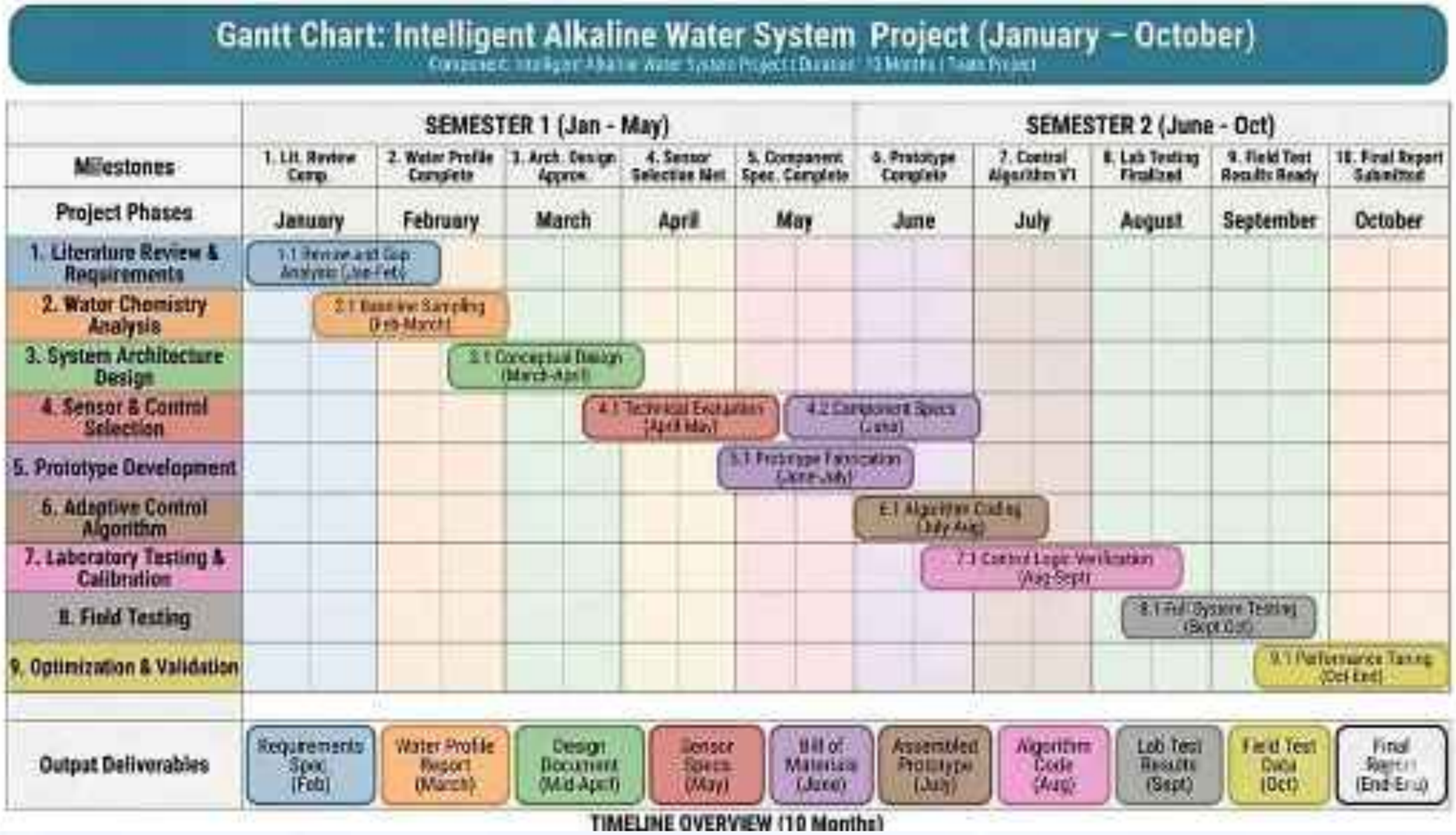
BUDGET & JUSTIFICATION



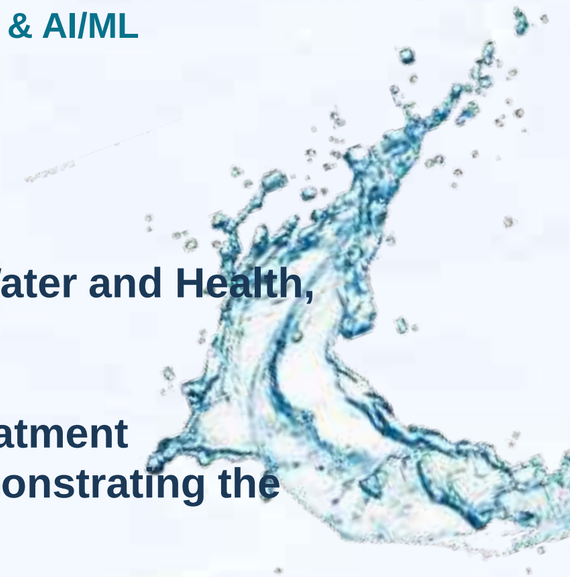
Item	Component/Description	Estimated Cost (LKR)	Justification
1. Dosing Mechanism	3x Mini Peristaltic Pumps, Silicon Tubing, Check Valves	3,000	Precision pumps are essential for accurate mineral dosing. Using three pumps allows for multi-mineral control.
2. Fluidics & Housing	3D Printed Flow Cell/Mixing Chamber, Mineral Solution Reservoirs (3x), Tubing, Fittings	4,500	For containing mineral solutions and ensuring proper mixing and sensor placement.
3. Mineral Solutions	Food-grade Calcium, Magnesium, Potassium compounds for creating concentrated solutions	3,000	The "consumables" for creating the alkaline water, needed for prototyping and testing.
4. Misc. Hardware	Breadboard, Jumper Wires, Motor Drivers/Relays, PCB (optional)	2,000	For assembling and connecting all electronic components.
TOTAL ESTIMATED BUDGET		**12,500**	

GANTT CHART




Gantt Chart: Intelligent Alkaline Water System Project (January – October)
SEMESTER 1 (Jan - May)
SEMESTER 2 (June - Oct)
Milestones
1. Lit. Review Comp
2. Water Profile Complete
3. Arch. Design Approv
4. Sensor Selection Met
5. Component Spec. Complete
6. Prototype Complete
7. Control Algorithm V1
8. Lab Testing Finalized
9. Field Test Results Ready
10. Final Report Submitted
Project Phases
January
February
March
April
May
June
July
August
September
October
1. Literature Review & Requirements
2. Water Chemistry Analysis
3. System Architecture Design
4. Sensor & Control Selection
5. Prototype Development
6. Adaptive Control Algorithm
7. Laboratory Testing & Calibration
8. Field Testing
9. Optimization & Validation
2.1 Baseline Sampling (Feb-March)
3.1 Conceptual Design (March-April)
4.1 Technical Evaluation (April-May)
4.2 Component Specs (June)
5.1 Prototype Fabrication (June-July)
6.1 Algorithm Coding (July-Aug)
7.1 Control Logic Verification (Aug-Sept)
8.1 Full System Testing (Sept-Oct)
9.1 Performance Tuning (Oct End)
Output Deliverables
Requirements Spec (Feb)
Water Profile Report (March)
Design Document (Mid-April)
Sensor Specs (May)
Bill of Materials (June)
Assembled Prototype (July)
Algorithm Code (Aug)
Lab Test Results (Sept)
Field Test Data (Oct)
Final Report (End-Oct)
TIMELINE OVERVIEW (10 Months)

REFERENCES



[1] Heil, A. (2012). Properties and Potential Health Benefits of Alkaline Water. Journal of Water and Health, 10(3), 112-118. (Provides the health context and rationale).

[2] Chen, L., et al. (2021). AI-Enabled Real-Time Mineral Balancing for Industrial Water Treatment Processes. IEEE Transactions on Industrial Informatics, 17(8), 5600-5609. (Key paper demonstrating the feasibility of ML for dosing control, albeit in an industrial context).

[3] Kumar, S., & Sharma, R. (2019). IoT-Based Water Quality Monitoring System: A Review. International Journal of Advanced Research in Computer Science, 10(4), 45-52. (Background on IoT sensing).

[4] Singh, P., & Patel, A. (2020). Smart Water Purification Using Machine Learning Techniques. Proceedings of the International Conference on Smart Systems and IoT. (General overview of ML in water purification).

[5] World Health Organization (WHO). (2017). Guidelines for Drinking-water Quality (4th ed.). WHO. (For context on safe pH and mineral levels in drinking water).



DEPENDENCIES

- Requires pre-treated water from Wijesingha's distiller or other purification stage.
- Depends on De Silva's water classification for initial water chemistry.

COMMON COMPONENTS

- Shares cloud backend and database with other subsystems.
- Integrated with unified dashboard for centralized monitoring and control.



IT22193940 | R.M.N.A.RATHNAYAKE

WT BOT – AUTONOMOUS WATER TESTING ROBOT

Novelty - A low-cost, smart robot boat that finds water pollution instantly and takes action on the spot. Normal robots just collect data and store it for later. But the WT Bot checks water quality in real-time, and the moment it detects something wrong (like harmful chemicals), it automatically collects a water sample at that exact location, marks it with GPS coordinates, and sends an instant alert to a cloud dashboard. This means pollution can be stopped immediately instead of waiting days or weeks for lab test results.

PROBLEM



Current methods are slow, manual, and reactive. Relying on physical samples and lab analysis, they provide sparse data that fails to detect transient pollution events like industrial spills. Without real-time monitoring or early warnings, we only confirm contamination after the fact. This leads to delayed decisions, poor environmental management, and puts personnel at risk.

SOLUTION

- Mobility - autonomous USV captures high-resolution spatial data;
- Accessibility - real-time cloud transmission and dashboard visualization;
- Intelligence - on-edge AI detects anomalies, triggers alerts, and enables adaptive responses for proactive management.

OBJECTIVES



To design, develop, and validate an Autonomous Water Testing Robot (WT Bot) that performs real-time, geo-referenced water quality monitoring across inland water bodies. This system offers a safer, more efficient, and data-rich alternative to manual sampling, enabling proactive environmental management.

SUB OBJECTIVES

- Platform: Durable, waterproof catamaran hull.
- Sensors: pH, turbidity, temperature, DO via microcontroller.
- Navigation: GPS waypoint following with obstacle avoidance.
- Telemetry: 4G/LoRa for real-time cloud data transmission.
- Visualization: Web dashboard for mapping, graphing, and alerts.
- Validation: Pool and field trials for accuracy and endurance.

METHODOLOGY



Application of Domain Knowledge

- Robotics: Terrain navigation & obstacle avoidance.
- AI: On-device anomaly detection (TinyML).
- Env. Science: Standard sampling protocols.
- IoT: LoRa/GSM data transmission.
- Geospatial: GPS location mapping.

Latest Technologies

- Hardware: Robotic chassis, GPS module, Arduino/RPi, sensors (pH, TDS, Turbidity, Temperature).
- Software: ROS (Robot Operating System) for control, TinyML frameworks (TensorFlow Lite) for on-bot anomaly detection, LoRa/GSM for data transmission.

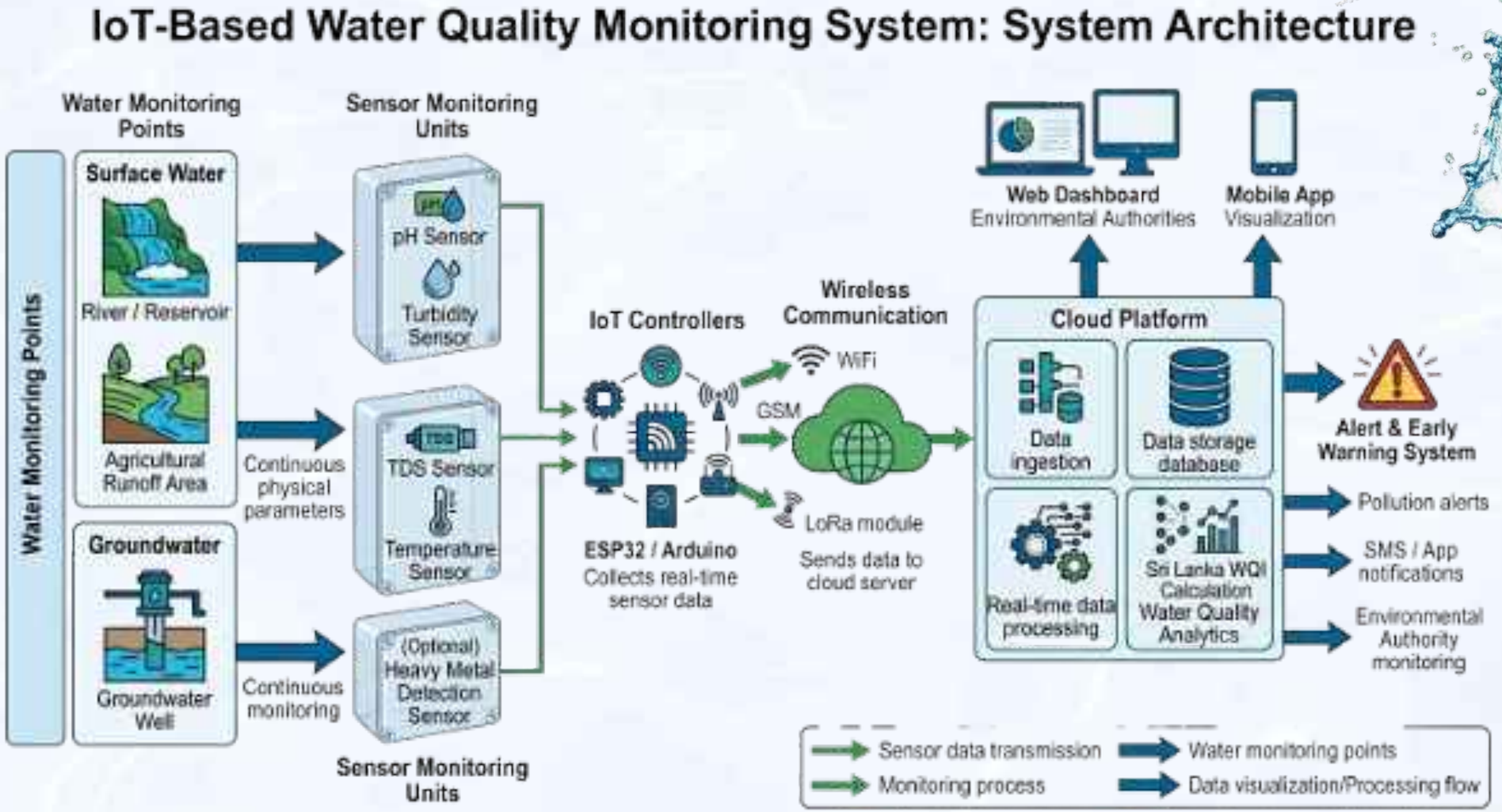
Validation & Evaluation

- Test navigation accuracy and obstacle avoidance.
- Validate sensor data accuracy against lab-controlled samples.
- Measure battery life and data transmission reliability.

Data & Ethics

- Data: Sensor data from field tests in controlled environments (university premises).
- Ethics: Clearance required for any testing in natural ecosystems to ensure minimal environmental disturbance.

SYSTEM DIAGRAM





DEPENDENCIES

- Provides external water quality data to retrain De Silva's classification model.
- Field testing requires environmental clearance.

COMMON COMPONENTS

- Shares cloud backend and database with mobile app and other subsystems.
- Data integrated into unified dashboard for real-time visualization and alerts.



FUNCTIONAL REQUIREMENTS

- Measure pH, turbidity, temperature, and dissolved oxygen
- Navigate autonomously to GPS waypoints
- Avoid obstacles using proximity sensors
- Transmit geo-tagged data to cloud every 10 seconds
- Log data locally on SD card
- Display real-time position and historical paths on dashboard
- Generate time-series graphs for each parameter
- Send alerts when user-defined thresholds are exceeded

NON-FUNCTIONAL REQUIREMENTS

- Waterproof (IP64 or higher)
- 2–3 hours battery endurance
- Sensor accuracy within ±5% of professional meters
- Intuitive, easy-to-use dashboard
- Modular design for easy field replacement
- Manual override for emergency retrieval

BUDGET & JUSTIFICATION



Item	Component/Description	Estimated Cost (LKR)	Justification
1. Core Electronics	Microcontroller (Raspberry Pi 4/Arduino Mega), Power Distribution Board, Cables, Connectors	1,000	The central processing unit: a Pi offers more scope for advanced features.
2. Water Quality Sensors	pH Electrode & Circuit, Turbidity Sensor, DS18B20 Temp Sensor, Dissolved Oxygen Probe	3,500	These are the primary data collection instruments. Quality sensors are critical for accuracy. DO probe is the most expensive.
3. Power System	12V Li-Po Battery (High Capacity), Battery Charger, Voltage Regulators	5,500	Provides operational endurance. A quality Li-Po battery is required for longer missions.
4. Hardware components	SSD	4,500	Covers recurring costs for cloud services and miscellaneous hardware needs.
TOTAL ESTIMATED BUDGET		**14,500**	

GANTT CHART



Gantt Chart: Continuous Water Quality Monitoring Project (January – October)

Component: Continuous Water Quality Monitoring Project | Duration: 10 Months | Team Project

Milestones	SEMESTER 1 (Jan - May)					SEMESTER 2 (June - Oct)				
	1. Project Launch	2. Site Survey Complete	3. Equipment Order Met	4. Proto. Assembled	5. Proto. Tested	6. Pilot Network Live	7. Platform Live	8. Network Fully Deployed	9. Training Complete	10. Final Report Submitted
Project Phases	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**
1. Planning & Requirements	1.1 Review and Gap Analysis (Jan-Feb)									
2. System Design		2.1 Conceptual & Detailed Design (Feb-March)								
3. Procurement			3.1 Component Sourcing (March-April)							
4. Prototype Development				4.1 Sensor Integration & Fabrication (April-May)						
5. Pilot Deployment					5.1 Field Site Prep & Initial Install (May-June)					
6. Data Platform Development						6.1 Platform Development & Initial Tests (June-July)				
7. Testing & Calibration							7.1 Network Testing & Calibration (July-August)			
8. Training & Capacity Building								8.1 User & Stakeholder Training (August-September)		
9. Evaluation & Reporting								9.1 Performance Evaluation & Final Reporting (September-October)		
Output Deliverables	Requirements Spec (Feb)	System Design Document (March)	Procurement Report (April)	Assembled Prototype (May)	Pilot Deploy Specs (June)	Data Platform Report (July)	Test & Calib Report (August)	Training Materials (Sept)	User Guide (Oct)	Final Evaluation Report (Oct - End)

TIMELINE OVERVIEW (10 Months)

REFERENCES



[1] Chapman, D. V. (Ed.). (1996). Water Quality Assessments: A Guide to the Use of Biota, Sediments and Water in Environmental Monitoring (2nd ed.). E & FN Spon. (A foundational text on traditional monitoring methods).

[2] Dunbabin, M., & Marques, L. (2012). Robots for Environmental Monitoring: Significant Advancements and Applications. IEEE Robotics & Automation Magazine, 19(1), 24-39. (Key review paper on the use of robotics in this domain).

[3] Giardino, C., et al. (2019). Imaging Spectrometry of Inland Water Quality: From Envisat to Sentinel-3. Surveys in Geophysics, 40(3), 461-485. (Covers remote sensing techniques).

[4] Kumar, R., & Singh, P. (2021). IoT Based Water Quality Monitoring System: A Review. Materials Today: Proceedings. (Provides an overview of static IoT monitoring systems).

[5] Rahman, M., et al. (2021). Design and Development of an Autonomous Surface Vehicle for Water Quality Monitoring. Proceedings of the International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), IEEE. (A direct example of a similar USV project).



IT22211750 | DE SILVA W.N.N.

DIFFERENT TYPES OF WATER IDENTIFIER – ML & AI

Novelty - The use of machine learning to automatically identify water types using combined multi-sensor data. Traditional water testing tools only display simple parameters values which still require human interpretation. In contrast, the proposed system analyzes multiple water quality parameters together and uses machine learning to intelligently classify the water type.

PROBLEM



- Water quality testing traditionally depends on laboratory analysis and manual chemical testing.
- These methods are time-consuming, expensive, and unsuitable for continuous monitoring.
- Existing IoT water monitoring systems mainly provide threshold-based alerts.
- However, they cannot intelligently classify different types of water based on sensor data.

RESEARCH GAP

- Current systems detect parameter anomalies, but they do not identify the actual water category.
- There is limited integration between sensor-based monitoring systems and machine learning classification models for water type identification.

MAIN OBJECTIVE



Develop an AI-based intelligent system capable of automatically identifying different water types using multi-sensor data and machine learning algorithms.

SUB OBJECTIVES

- Design a multi-sensor data acquisition system using water quality sensors.
- Create a dataset of different water types using sensor readings.
- Train machine learning models to classify water types.
- Integrate the trained ML model with real-time sensor data.
- Provide a user interface for monitoring and visualization.

METHODOLOGY



Domains

- **Knowledge Domain : Artificial Intelligence**
- **Subdomains : Machine Learning, Environmental Data Analysis, Sensor-Based Monitoring Systems**

Latest Technologies

- **Hardware: ESP32 Microcontroller, pH Sensor,TDS Sensor,Turbidity Sensor**
- **Software: Python, Pandas, NumPy**

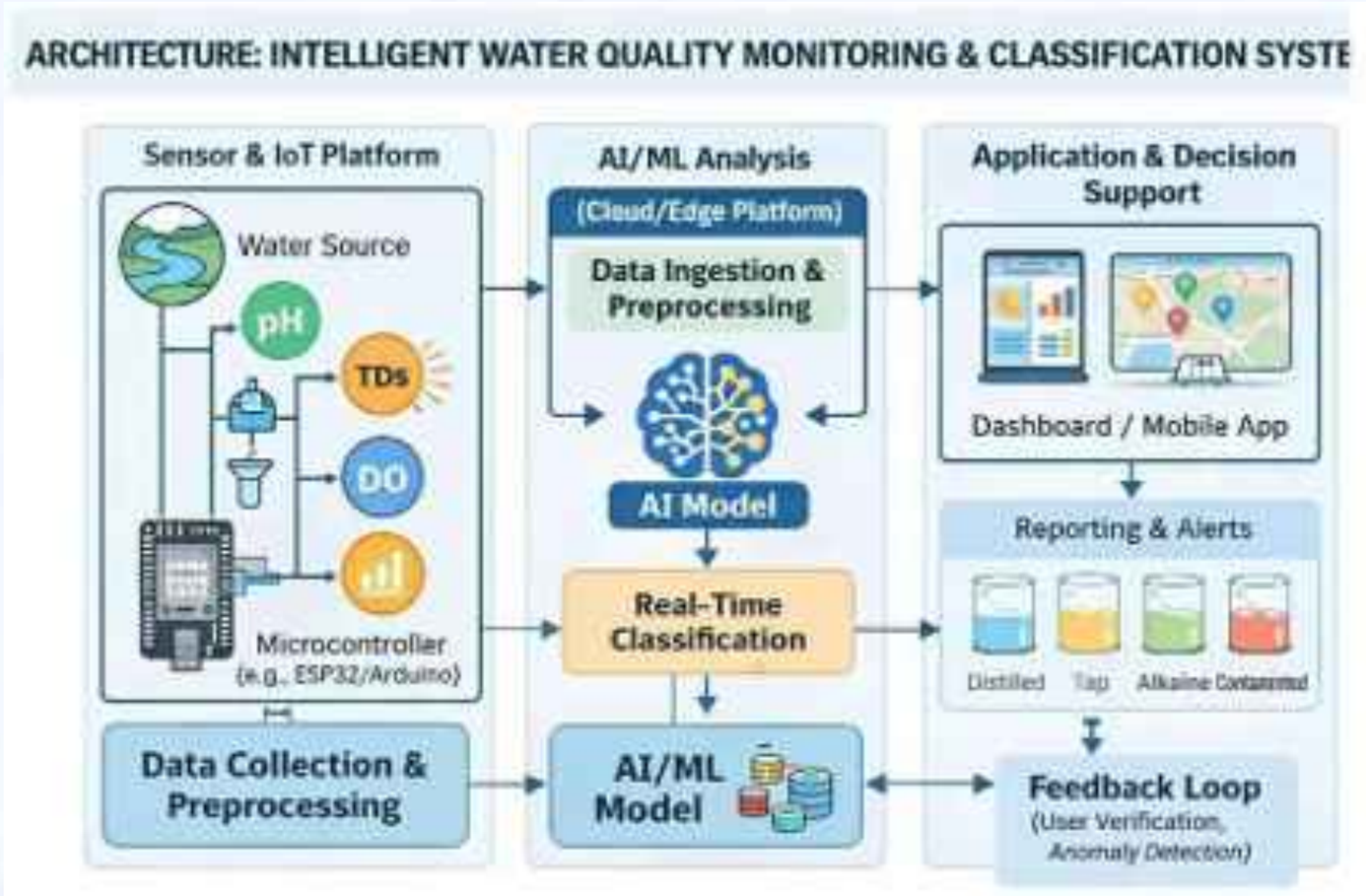
Validation & Evaluation

- **Model Evaluation :Dataset Split ,Training – 70% , Validation – 15% , Testing – 15% ,Evaluation Metrics , Accuracy , Precision**

Data Availability

- Dataset generated from controlled water samples
- No personal or sensitive data involved
- Data used only for research purposes
- Research conducted according to academic ethical guidelines

SYSTEM DIAGRAM





DEPENDENCIES

Hardware Dependencies

- Water quality sensor modules
- ESP32 microcontroller

Software Dependencies

- Data preprocessing module
- Machine learning classification model
- Visualization dashboard

COMMON COMPONENTS

- Sensor data acquisition system
- Dataset generation module
- User interface dashboard



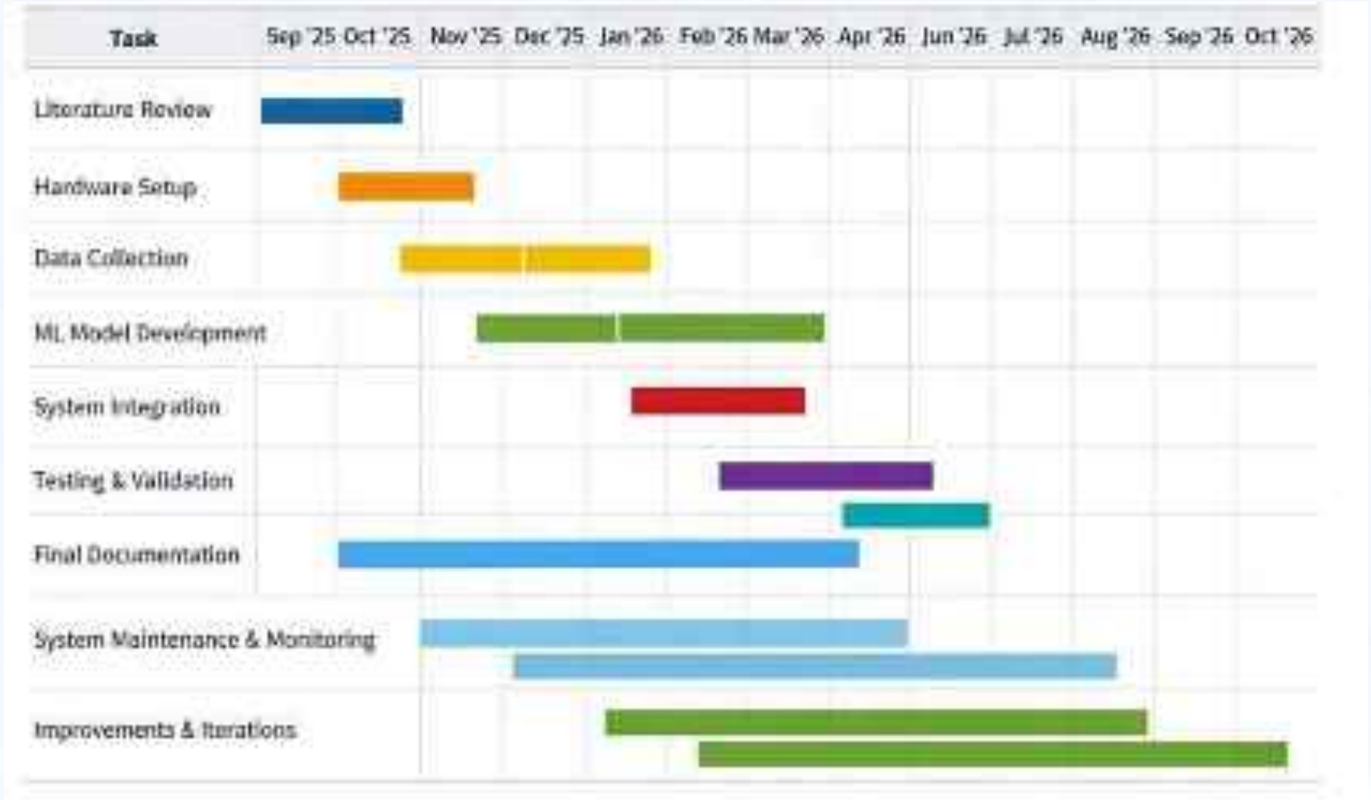
FUNCTIONAL REQUIREMENTS

- Collect water quality parameters (pH, turbidity, TDS, conductivity, temperature).
- Preprocess and normalize sensor data for analysis.
- Apply machine learning model to classify water types.
- Display identified water type and sensor readings on a dashboard.
- Store collected data for future analysis and model training.

NON-FUNCTIONAL REQUIREMENTS

- Accuracy: System should provide reliable water type classification.
- Performance: Results should be generated within a few seconds.
- Usability: Dashboard should be simple and easy to understand.
- Reliability: System should process sensor data consistently.
- Scalability: System should support additional sensors in the future.

GANTT CHART


Task
Sep '25
Oct '25
Nov '25
Dec '25
Jan '26
Feb '26
Mar '26
Apr '26
Jun '26
Jul '26
Aug '26
Sep '26
Oct '26
Literature Review
Hardware Setup
Data Collection
ML Model Development
System Integration
Testing & Validation
Final Documentation
System Maintenance & Monitoring
Improvements & Iterations

BUDGET AND JUSTIFICATION



Component	Quantity	Unit Price (LKR)	Total Cost
ESP32 Microcontroller	1	2,500	2,500
pH Sensor Module	1	1,500	1,500
TDS Sensor	1	1,000	1,000
Turbidity Sensor	1	500	500
Breadboard & Wires	1	1,000	1,000
Power Supply	1	5000	5000
Miscellaneous Components	-	2,000	2,000

The proposed system aims to develop a cost-effective water identification device. All components were selected based on affordability, availability, and compatibility with microcontroller platforms.

DATA SETS



1. Water Potability Dataset – Kaggle

- **Contains water quality parameters such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, and turbidity.**
- **Used for training and testing machine learning classification models.**

2. Water Quality Dataset – UCI Machine Learning Repository

- **Provides environmental water quality parameters including pH, turbidity, dissolved oxygen, and temperature.**
- **Used to support water quality analysis and model validation.**

3. Sensor-Collected Dataset (Proposed System)

- **Data collected using sensors measuring pH, turbidity, TDS, conductivity, and temperature.**
- **Used for real-time water type classification and model evaluation.**

REFERENCES



[1] APHA. (2017). Standard Methods for the Examination of Water and Wastewater (23rd ed.). American Public Health Association. (The definitive guide to traditional water testing methods).

[2] Kumar, R., & Singh, P. (2021). IoT Based Smart Water Quality Monitoring System: A Review. Materials Today: Proceedings. (Overview of IoT systems in the field).

[3] Najah, A., et al. (2013). Application of Artificial Neural Networks for Water Quality Prediction. Neural Computing and Applications, 22(1), 187-201. (Example of using ANNs for predicting a specific parameter).

[4] Uddin, M. G., et al. (2020). Performance Analysis of Machine Learning Algorithms for Water Quality Index Prediction. Journal of Environmental Informatics, 36(2). (Key paper comparing ML algorithms for water quality tasks).

[5] Winquist, F., et al. (2005). An Electronic Tongue in the Food Industry. Sensors and Actuators B: Chemical, 111-112, 299-304. (Background on multi-sensor arrays and pattern recognition for liquid classification).

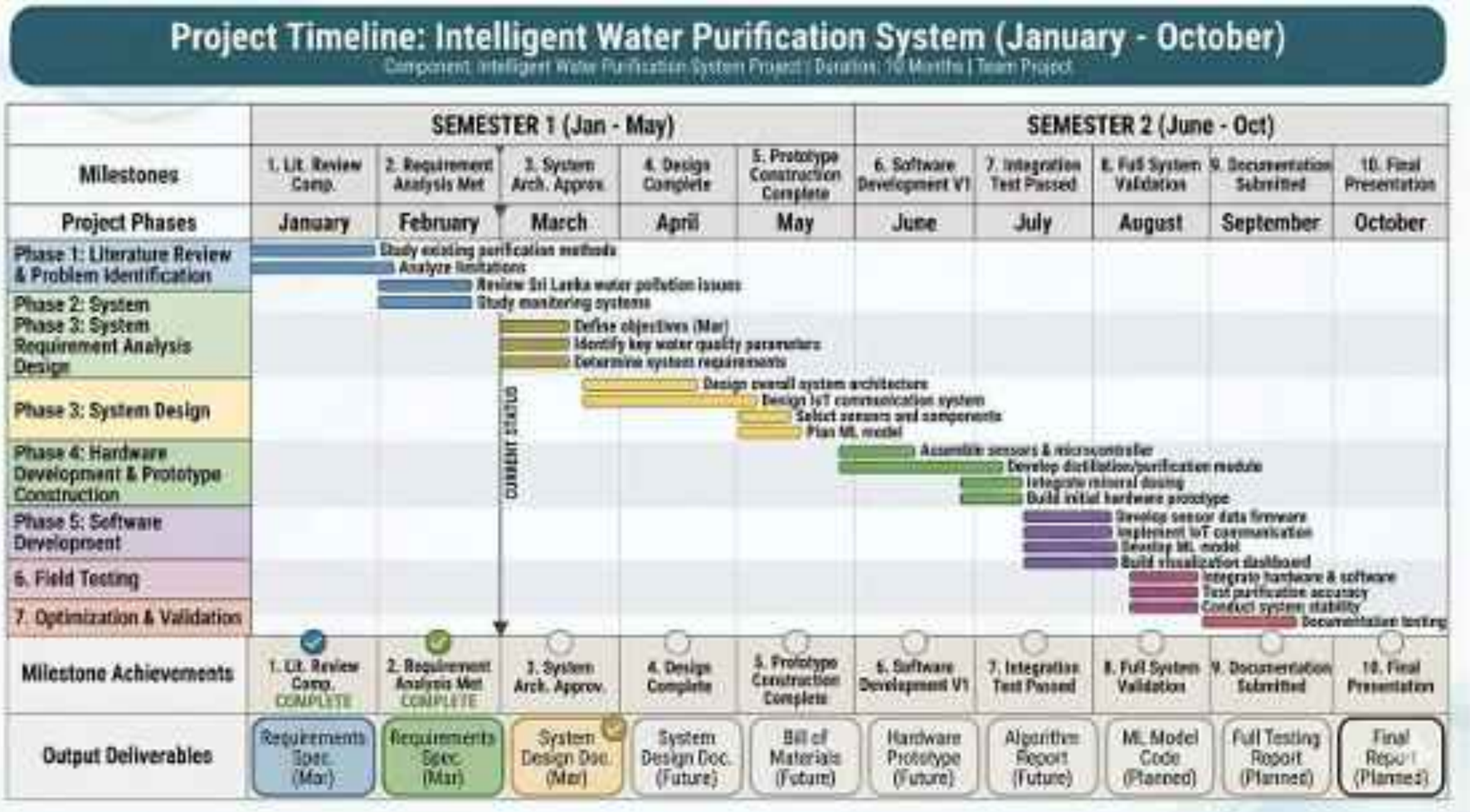
ABILITY OF COMMERCIALIZATION



- High potential as a premium health and wellness home appliance, with a B2B model for the WT Bot (government/environmental agencies).
- Investment/Cost:
- Labour: 4 Final Year Students.
- Material: Prototyping cost: LKR 200,000 (sensors, microcontrollers, pumps, chassis).
- Cost Recovery (Pricing):
- B2C Model: One-time sale of unit (LKR 100,000 - LKR 150,00) with a subscription for mineral refills and AI model updates.
- B2G/B2B Model: Service contract for WT Bot monitoring and data analytics.
- Target Market: Health-conscious households in urban areas; NGOs and government agencies in CKDu-affected regions.
- Value Proposition: "The only system that thinks before it purifies, ensuring your water is not just clean, but optimally healthy."
- Intellectual Property: Potential patent for the "AI-driven adaptive mineral balancing system" and the "integrated water identification and purification logic.

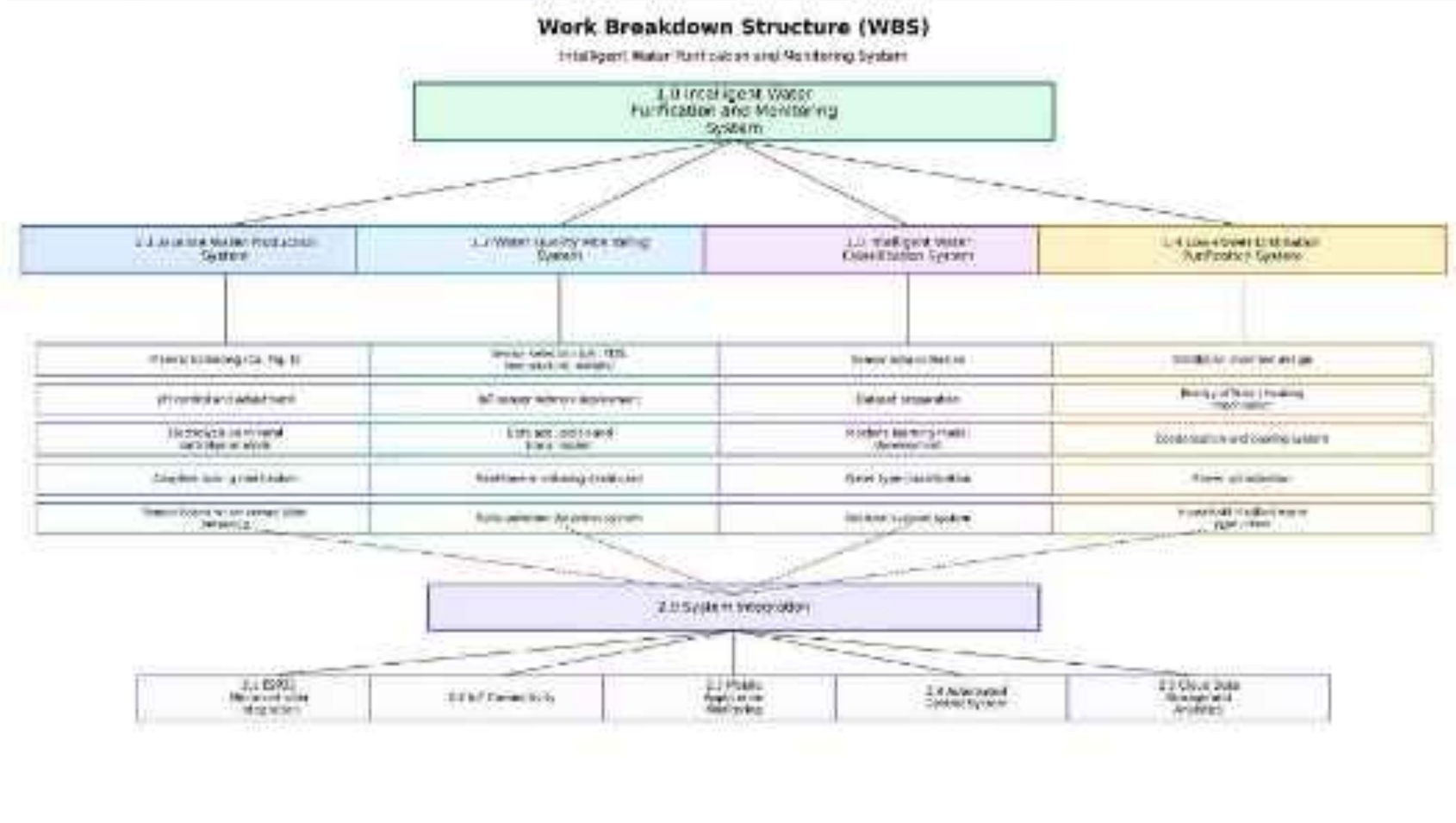
GANTT CHART




Project Timeline: Intelligent Water Purification System (January - October)
Component: Intelligent Water Purification System Project | Duration: 10 Months | Team Project
SEMESTER 1 (Jan - May)
SEMESTER 2 (June - Oct)
Milestones
1. Lit. Review Comp.
2. Requirement Analysis Met
3. System Arch. Approv.
4. Design Complete
5. Prototype Construction Complete
6. Software Development V1
7. Integration Test Passed
8. Full System Validation
9. Documentation Submitted
10. Final Presentation
Project Phases
January
February
March
April
May
June
July
August
September
October
Phase 1: Literature Review & Problem Identification
Study existing purification methods
Analyze limitations
Review Sri Lanka water pollution issues
Study monitoring systems
Phase 2: System Phase 3: System Requirement Analysis Design
Define objectives (Mar)
Identify key water quality parameters
Determine system requirements
Phase 3: System Design
Design overall system architecture
Design IoT communication system
Select sensors and components
Plan ML model
Phase 4: Hardware Development & Prototype Construction
Assemble sensors & microcontroller
Develop distillation/purification module
Integrate mineral dosing
Build initial hardware prototype
Phase 5: Software Development
Develop sensor data firmware
Implement IoT communication
Develop ML model
Build visualization dashboard
6. Field Testing
Integrate hardware & software
Test purification accuracy
7. Optimization & Validation
Conduct system stability
Documentation testing
CURRENT STATUS
Milestone Achievements
1. Lit. Review Comp. COMPLETE
2. Requirement Analysis Met COMPLETE
3. System Arch. Approv.
4. Design Complete
5. Prototype Construction Complete
6. Software Development V1
7. Integration Test Passed
8. Full System Validation
9. Documentation Submitted
10. Final Presentation
Output Deliverables
Requirements Spec (Mar)
Requirements Spec (Mar)
System Design Doc (Mar)
System Design Doc (Future)
Bill of Materials (Future)
Hardware Prototype (Future)
Algorithm Report (Future)
ML Model Code (Planned)
Full Testing Report (Planned)
Final Report (Planned)

WORK BREAKDOWN STRUCTURE





THANK YOU

ANY QUESTIONS ?

